Functional Annotation of Bovine α-Lactalbumin (LALBA, UniProt P00711)

Target: LALBA (synonym *ALACTA*) — α-lactalbumin; "Lactose synthase B protein"; Allergen Bos d 4 **Organism:** *Bos taurus* (bovine) **Family:** Glycosyl hydrolase 22 family (c-type lysozyme-like fold); domains Glyco_hydro_22 (IPR001916), Lactalbumin (IPR000545), Lysozyme-like domain superfamily (IPR023346); Pfam PF00062 (Lys)

Summary

Bovine **α-lactalbumin** (gene **LALBA**, synonym *ALACTA*; UniProt **P00711**; *Bos taurus*) is a small (~14.2 kDa), acidic, calcium-binding whey protein whose primary function is **not** enzymatic catalysis. Rather, it serves as the regulatory **"B protein" (specifier) subunit of lactose synthase** (EC 2.4.1.22). The catalytic partner is **β-1,4-galactosyltransferase-1 (B4GALT1)**, which on its own transfers galactose from UDP-galactose onto *N*-acetylglucosamine (GlcNAc) to make *N*-acetyllactosamine. α-Lactalbumin forms a 1:1 heterodimer with B4GALT1 and **reprograms its acceptor specificity**, redirecting galactose transfer onto **free glucose** to synthesize **lactose** (Gal-β-1,4-Glc), the principal sugar of milk. This single biochemical switch is the molecular basis of copious milk-carbohydrate production in lactating mammals.

Structurally, α-lactalbumin is a **catalytically inactive homolog of c-type lysozyme** (glycosyl hydrolase family 22), arising from a gene duplication 300–400 million years ago. It shares ~40% sequence identity and a near-identical two-lobe fold with lysozyme, but specific amino-acid substitutions abolished the ancestral bacteriolytic (glycosidase) activity and instead conferred the specifier role. It carries a single strong Ca^{2+}-binding site (plus secondary Zn^{2+}/Mn^{2+} sites) and is the textbook model of the **molten globule** partially-folded state. The protein acts in the **Golgi lumen of lactating mammary epithelial cells**, co-localizing with Golgi markers alongside its membrane-bound catalytic partner, and is subsequently **secreted into milk** as one of the most abundant whey proteins (up to ~20% of total milk protein). In milk it is clinically important as the major whey allergen **Bos d 4**.

This report consolidates ten confirmed findings drawn from 46 reviewed papers spanning classical enzymology, high-resolution crystallography, molecular evolution, protein-folding biophysics, metal-binding studies, mammary cell biology, and allergy immunology. Together they provide a coherent, mechanistically precise answer: α-lactalbumin is the conformational "switch" protein that converts a general galactosyltransferase into a lactose synthase.

Key Findings

Finding 1 — Primary function: the regulatory "specifier" subunit of lactose synthase (EC 2.4.1.22)

The defining function of α-lactalbumin is that of a **non-catalytic regulatory component** of the heterodimeric enzyme **lactose synthase**. B4GALT1 (the "A protein") is a β-1,4-galactosyltransferase that, acting alone, transfers galactose from the donor UDP-galactose onto the acceptor GlcNAc, producing *N*-acetyllactosamine. When α-lactalbumin binds B4GALT1 in a 1:1 complex, it **alters the acceptor specificity** so the enzyme instead transfers galactose onto free **glucose**, producing lactose — milk's principal carbohydrate (PMID: 11419947).

Classical enzymology established the mechanism of this specificity change. α-Lactalbumin acts through **competitive, mutually exclusive binding** at a site on the galactosyltransferase close to the monosaccharide-binding site, competing with GlcNAc-type acceptors while simultaneously **increasing the enzyme's affinity for glucose** (PMID: 3922406). A review of the native protein's role summarizes that this substrate-specifier function "defines a function essential for the survival of lactating mammals" (PMID: 23130929).

> "LA promotes the binding of glucose (Glc) to beta4Gal-T1, thereby altering its sugar acceptor specificity from N-acetylglucosamine (GlcNAc) to glucose, which enables LS to synthesize lactose, the major carbohydrate component of milk." — PMID: 11419947
>
> "alpha-Lactalbumin must bind to a region on galactosyltransferase in close proximity to the monosaccharide binding site and exert its substrate-specifying action through competitive and mutually exclusive binding with the N-acetylglucosamine analogues accompanied by an increased affinity for glucose" — PMID: 3922406

Reaction catalyzed by the complex:

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                    α-lactalbumin (specifier)
UDP-galactose + D-glucose  ──────────────────────►  lactose (Gal-β-1,4-Glc) + UDP
                    B4GALT1 (catalytic) + Mn²⁺
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Finding 2 — Structural mechanism: α-lactalbumin holds glucose and stabilizes the closed conformation of B4GALT1

High-resolution (~2 Å) crystal structures of the lactose synthase complex resolved exactly how α-lactalbumin works. Substrate binding drives B4GALT1 from an **open to a closed conformation**, involving a large movement of a flexible loop (residues ~345–365) that repositions His347 for metal coordination and creates the composite binding surface for α-lactalbumin and the acceptor sugar (PMID: 11419947). α-Lactalbumin binds at the enzyme's C-terminal flexible region and **holds glucose by hydrogen-bonding to its O1 hydroxyl** in the acceptor site (PMID: 11419947).

Critically, the catalytically competent **closed conformation** of B4GALT1, when bound to glucose or less-preferred substrates, is observed **only in the presence of α-lactalbumin** (PMID: 12854954). α-Lactalbumin also enhances the enzyme's otherwise negligible intrinsic glucosyltransferase activity ~25–30-fold (PMID: 11485999). The flexible **C-terminal portion of α-lactalbumin** is the principal interaction surface for the catalytic component (PMID: 8805552).

> "the role of LA is to hold Glc by hydrogen bonding with the O-1 hydroxyl group in the acceptor-binding site on beta4Gal-T1" — PMID: 11419947
>
> "The closed conformation induced by the sugar acceptors or the less preferred donor substrates has been observed only when complexed with LA" — PMID: 12854954
>
> "LA's proposed interaction site for the catalytic component of the lactose synthase complex is primarily located in the flexible C-terminal portion of the molecule" — PMID: 8805552

Finding 3 — Evolution and fold: a catalytically inactive c-type lysozyme homolog (glycosyl hydrolase family 22)

α-Lactalbumin is an evolutionary descendant of **lysozyme C**, having arisen from a gene duplication **300–400 million years ago** (PMID: 9307874). It retains ~40% sequence identity and a near-identical backbone fold — two lobes comprising a large α-helical domain and a small β-

sheet domain, with a bound Ca^{2+} — confirmed by crystallography (PMID: 1920433). Despite this structural conservation, **specific amino-acid substitutions abolished the ancestral glycosidase/bacteriolytic activity** and conferred the new specifier function (PMID: 9307874).

A single tryptophan residue is functionally essential: chemical nitration or modification of one tryptophan completely abolishes specifier activity, even though the protein otherwise remains folded (PMID: 812700).

> "Lysozyme C gene also gave rise, after gene duplication 300 to 400 million years ago, to a gene that currently codes for alpha-lactalbumin" — PMID: 9307874
>
> "Specific amino acid substitutions in alpha-lactalbumin account for the loss of the enzyme activity of lysozyme and the acquisition of the features necessary for its role in lactose synthesis" — PMID: 9307874
>
> "The striking structural resemblance between alpha-lactalbumins and C-type lysozymes emphasizes the homologous evolutionary relationship between these two classes of proteins" — PMID: 1920433

This evolutionary history explains the UniProt family assignment to the **glycosyl hydrolase 22 family** and the lysozyme-like domain signatures (Glyco_hydro_22, Lactalbumin, Lysozyme-like_dom_sf, PF00062) — α-lactalbumin inherits the structural scaffold of a glycosidase but has lost catalysis.

Finding 4 — A calcium-binding protein and the classic molten-globule model

α-Lactalbumin is a small (Mr ~14,200), simple, acidic (pI 4–5) **Ca^{2+}-binding protein** that can constitute up to **~20% of total milk protein** (PMID: 26956441). Its two domains are joined by a **calcium-binding loop**. Removal of the single tightly bound Ca^{2+} (for example at acidic pH) converts the protein into a **molten globule** — a compact but partially folded state lacking fixed tertiary structure — making α-lactalbumin the canonical textbook model of this intermediate (PMID: 26956441, PMID: 36043462).

Metal-ion cofactor binding **accelerates and organizes folding**, lowering the energy barrier between the molten globule and transition states (PMID: 16522796), and the bound calcium strongly stabilizes the native structure and governs the folding pathway (PMID: 20080106).

> "a small (Mr 14 200), simple, acidic (pI 4-5), Ca(2+)-binding protein that might constitute up to 20% of total milk protein" — PMID: 26956441
>
> "protein has two domains, a large α-helical domain and a small β-sheet domain connected by a calcium binding loop" — PMID: 26956441
>
> "α-lactalbumin is well-known as a classic example of the molten globule state" — PMID: 26956441

Finding 5 — Moonlighting activity: molten-globule α-lactalbumin + oleic acid forms tumoricidal HAMLET/BAMLET

A striking conformation-dependent "moonlighting" activity emerges when calcium is removed and α-lactalbumin adopts its molten-globule form: complexation with **oleic acid** produces **HAMLET** (Human Alpha-lactalbumin Made LEthal to Tumor cells) or, for the bovine protein, **BAMLET**, a protein–lipid complex that **selectively kills tumor cells** (PMID: 23130929, PMID: 22016817). The fatty acid both stabilizes the altered fold and co-activates death pathways; cellular targets span the plasma membrane, cytoskeleton, mitochondria, proteasomes, lysosomes, and nuclei, with therapeutic effects demonstrated in papillomas, bladder cancer, and glioblastoma models (PMID: 23130929).

> "α-lactalbumin acquires tumoricidal activity after partial unfolding and binding to oleic acid" — PMID: 23130929
>
> "When calcium is removed from α-LA, it adopts a molten globule form, and this form, interestingly, when complexed with oleic acid (OA) acquires tumoricidal activity" — PMID: 22016817

While scientifically important, this is a secondary (non-physiological for milk) activity and is noted here for completeness rather than as the protein's primary biological role.

Finding 6 — Subcellular localization: acts in the Golgi of lactating mammary epithelial cells, then secreted into milk

Lactose is synthesized in the **Golgi apparatus of mammary epithelial cells** by lactose synthase, from glucose and UDP-galactose (PMID: 31020623). In mammary tissue, α-lactalbumin **co-localizes with Golgi markers** (the 58K protein / GM130), consistent with the

lactose synthase complex residing on the luminal face of the *trans*-Golgi, where the catalytic partner B4GALT1 is anchored as a type-II Golgi membrane protein. Glucose is supplied to the Golgi lumen by glucose transporters (GLUT1/GLUT8), which partially co-localize with α-lactalbumin in mammary tissue (PMID: 31020623).

α-Lactalbumin is itself a **secreted protein** — synthesized with a signal peptide (UniProt flags it as a precursor), carried through the secretory pathway, and released into milk as a soluble whey protein (PMID: 26956441).

> "Lactose, the principal carbohydrate of the milk, is synthetized in the Golgi of mammary epithelial cells by lactose synthase from glucose and UPD galactose" — PMID: 31020623
>
> "Endogenous GLUT8 partially co-localized with 58 K protein and α-lactalbumin in mammary tissue" — PMID: 31020623

Finding 7 — Bos d 4: a major cow's-milk whey allergen with defined IgE epitopes

In milk, bovine α-lactalbumin is clinically significant as the allergen **Bos d 4**, one of the major whey allergens in cow's-milk allergy — a condition affecting roughly **8% of the pediatric population** (PMID: 37406758, PMID: 37641259). Both **linear and conformational IgE/IgG epitopes** have been mapped on native Bos d 4 using patient sera, phage-display mimotopes, and alanine-scanning mutagenesis — including a linear epitope at residues 11–26 and several conformational epitopes — work that informs the design of hypoallergenic dairy products (PMID: 27273451, PMID: 37641259, PMID: 27603546).

> "α-Lactalbumin (α-LA) and β-lactoglobulin (β-LG) from whey protein make up a relatively high proportion of milk proteins and have received widespread attention as major allergens in milk" — PMID: 37641259
>
> "Alpha-lactalbumin (ALA) is one of the major allergens in cow's milk" — PMID: 27273451

Finding 8 — Physiological consequence: lactose is the osmotic regulator of milk secretion

The functional output enabled by α-lactalbumin — lactose — is the **major osmotic regulator of lactation**. Because lactose is osmotically active and is secreted into the alveolar lumen, it draws water into the milk and thereby sets **milk volume** (PMID: 12216958). Genetic manipulation of lactose synthesis (including of α-lactalbumin) in transgenic animals alters milk carbohydrate composition and milk fluidity/volume, confirming its rate-determining role (PMID: 12216958).

Crucially, while B4GALT1 is **widely expressed** across tissues, α-lactalbumin expression is **restricted to, and hormonally induced in, lactating mammary epithelium** (via prolactin/STAT5 signaling) (PMID: 21688337). α-Lactalbumin availability is therefore the **molecular switch** that turns on lactose — and thus copious milk — synthesis.

> "Lactose is the major sugar present in milk and an important osmotic regulator of lactation" — PMID: 12216958
>
> "A decrease of 39% and 23% in WAP and α-lactalbumin expression, respectively" — PMID: 21688337

Finding 9 — The "B protein" of lactose synthase and a general β4-glycosyltransferase acceptor-specificity modifier

Classical and structural work defines lactose synthase as a heterodimer in which the **"A protein" is the catalytic galactosyltransferase (B4GALT1)** and the **"B protein" is α-lactalbumin** (PMID: 168914, PMID: 1204653) — this is the origin of the UniProt alternate name "Lactose synthase B protein." A mechanistic review describes the lactose synthase system as a **molecular mechanical device** whose closed "conformation II" is required for α-lactalbumin binding and glucose selection (PMID: 11883930).

The modifier action is **general** and not restricted to the mammalian enzyme: α-lactalbumin can re-specify a homologous invertebrate β4-GalNAc-transferase from the snail *Lymnaea stagnalis*, inducing it to act on glucose with >100-fold increased efficiency to form the lactose analog GalNAc-β-1,4-Glc (PMID: 8816760). This confirms that α-lactalbumin's role is a transferable, intrinsic property of modulating β4-glycosyltransferase acceptor selection.

> "No lactose synthetase B protein (alpha-lactalbumin) has been discovered in the tumours" — PMID: 168914
>
> "Only in conformation II do Gal-T1 and LA form the LS complex, enabling Gal-T1 to choose the new substrate glucose" — PMID: 11883930
>
> "the snail beta 4-GalNAcT was induced by this protein to act on Glc with a > 100-fold increased efficiency, resulting in the formation of the lactose analog GalNAc beta 1-->4Glc" — PMID: 8816760

Finding 10 — Bovine metal-binding architecture: strong Ca^{2+} site plus distinct Zn^{2+} and secondary Ca^{2+}/Mn^{2+} sites near the functional surface

Beyond its single strong Ca^{2+} site, bovine α-lactalbumin carries **distinct zinc-binding sites** — one strong site plus weaker secondary sites. In the bovine protein, strong Zn^{2+} binding depends on **Glu1** (its substitution by Met abolishes strong Zn^{2+} binding), with a proposed site formed by **Glu1, Glu7, Asp11, and Asp37**; notably, sequence homology to human α-lactalbumin did **not** predict these subtle functional differences (PMID: 10813835).

A high-resolution structure additionally revealed a **secondary Ca^{2+} site ~7.9 Å from the primary site** that shares features with the Mn^{2+} site; its location on the charged functional surface led the authors to propose that these metal sites may participate in **formation of the lactose synthase complex** (PMID: 9537992). Secondary Zn^{2+} binding destabilizes the fold and promotes aggregation (PMID: 7945790).

> "substitution of Glu 1 by Met resulted in the disappearance of strong Zn(2+) binding. A proposed site involves Glu 1, Glu 7, Asp 11, and Asp 37" — PMID: 10813835
>
> "The proximity of the manganese and calcium binding region and the location of the functional site on one side of the charged surface of the alpha-lactalbumin molecule suggest that these binding sites might play a role in the formation of the lactose synthase complex" — PMID: 9537992

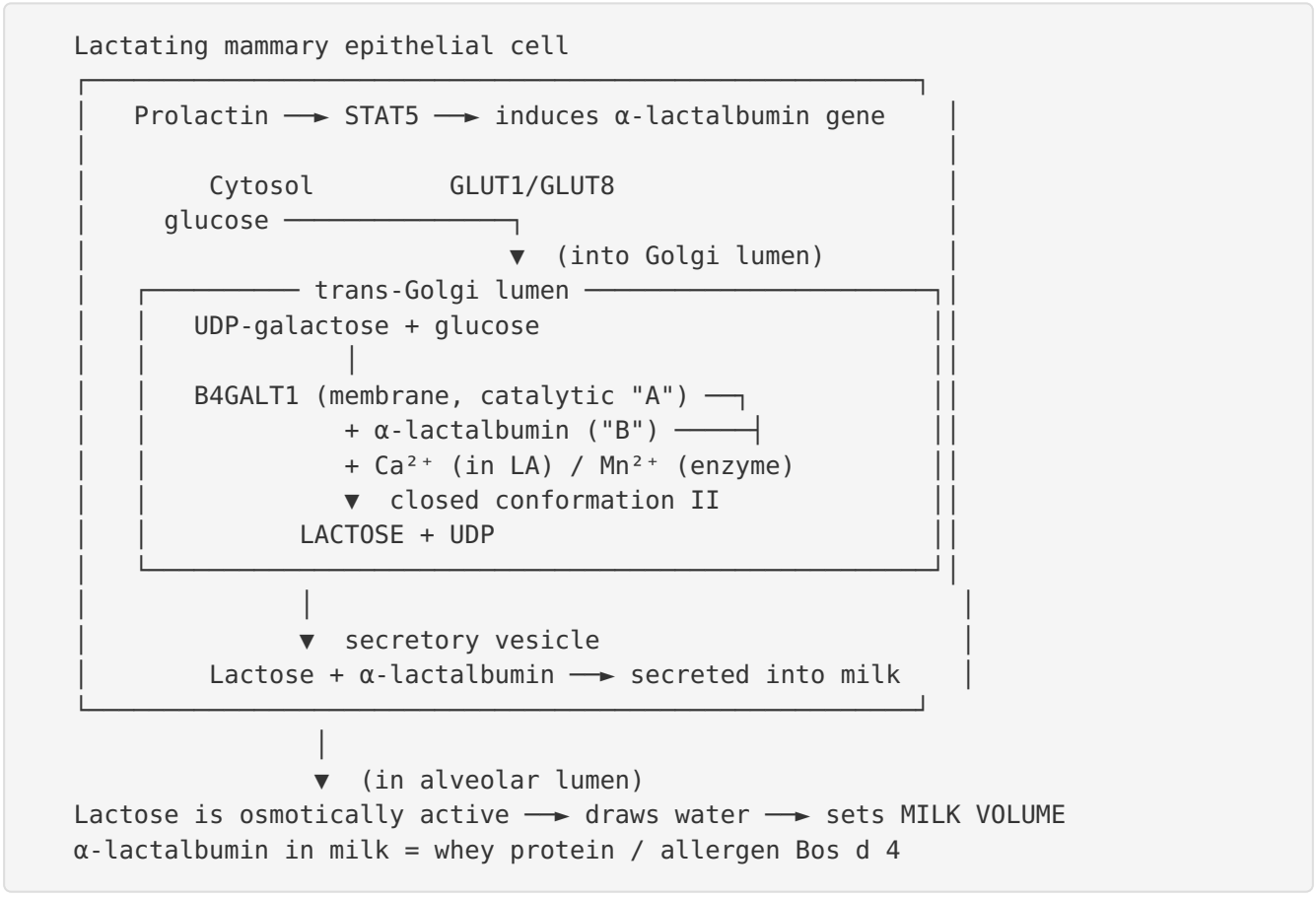
Mechanistic Model / Interpretation

The findings converge into a single, coherent mechanistic narrative: **α-lactalbumin is a conformational switch protein that converts a housekeeping glycosyltransferase into a lactose-synthesizing enzyme, restricted to lactating mammary tissue.**

The two enzymatic modes of B4GALT1

Component present	Acceptor substrate	Product	Biological context
B4GALT1 alone (ubiquitous)	*N*-acetylglucosamine (GlcNAc)	*N*-acetyllactosamine (Gal-β-1,4-GlcNAc)	Glycoprotein/glycolipid synthesis in all tissues
B4GALT1 + **α-lactalbumin**	**Glucose**	**Lactose** (Gal-β-1,4-Glc)	Milk-sugar synthesis in lactating mammary Golgi

Spatial and regulatory logic



Why this design is elegant

1. **Economy of parts.** Evolution did not invent a new catalytic enzyme for lactose. Instead, it repurposed a pre-existing, ubiquitous galactosyltransferase (B4GALT1) and built a small, catalytically dead regulatory protein (from a lysozyme gene) whose only job is to redirect the enzyme's acceptor specificity.
2. **Tissue specificity via the specifier.** Because B4GALT1 is everywhere but α-lactalbumin is hormonally restricted to lactating mammary tissue, lactose is made **only where and when** α-lactalbumin is present. The specifier protein is the control valve.
3. **Conformational coupling.** α-Lactalbumin works by stabilizing the enzyme's closed conformation and physically holding glucose in the acceptor site via a hydrogen bond to its O1 hydroxyl — a direct structural explanation for the specificity change, confirmed by multiple ~2 Å crystal structures.

4. **A lost-catalysis homolog repurposed.** The lysozyme-to-α-lactalbumin transition is a textbook case of neofunctionalization: the same fold, the same calcium-binding architecture, but specific substitutions that traded bacteriolytic glycosidase activity for protein–protein modulation.

The protein's prominent biophysical properties — molten-globule behavior, calcium dependence, metal-binding promiscuity — are secondary to, but mechanistically connected with, its primary role: calcium stabilizes the native fold required for productive complex formation, and the charged functional surface bearing secondary metal sites is implicated in assembling the lactose synthase complex.

Evidence Base

PMID	Study type	Contribution	Supports finding
11419947	Crystallography (~2 Å)	Lactose synthase structure; LA holds glucose O1; open→closed change	F1, F2
3922406	Classical enzymology/ kinetics	Competitive, mutually exclusive binding; raised glucose affinity	F1
23130929	Review (HAMLET)	Native specifier role "essential for lactating mammals"; tumoricidal moonlighting	F1, F5
12854954	Crystallography	Closed conformation observed only with LA	F2
11485999	Crystallography + kinetics	LA enhances Glc-T activity ~30×; UDP-Glc complex structure	F2
8805552	Crystallography (bovine, goat, guinea-pig)	Flexible C-terminal region is the enzyme-interaction surface	F2
9307874	Molecular evolution	LA from lysozyme-C duplication 300–400 Mya; substitutions abolish catalysis	F3
1920433	Crystallography (human LA, 1.7 Å)	Structural resemblance to C-type lysozyme; bound Ca^{2+}	F3, F4
812700	Chemical modification	Single Trp nitration abolishes specifier activity	F3
26956441	Review	Size, acidity, Ca^{2+} binding, ~20% of milk protein, molten globule	F4, F6
36043462	Simulation/biophysics	Molten-globule conformational dynamics	F4
16522796	Folding kinetics	Metal ions accelerate refolding	F4
20080106	CD/NMR folding	Ca^{2+} organizes folding pathway	F4
22016817	Biochemistry (HAMLET)	Ca^{2+} removal → molten globule + oleic acid → tumoricidal	F5
31020623	Cell biology	Lactose made in Golgi; LA co-localizes with Golgi 58K/GLUT8	F6

PMID	Study type	Contribution	Supports finding
37406758	Review	Cow's-milk allergy affects ~8% of children	F7
37641259	Phage display/epitope mapping	LA major whey allergen; epitope residues	F7
27273451	Epitope mapping	IgE/IgG epitopes on native Bos d 4	F7
27603546	Alanine scanning	Critical epitope residues	F7
12216958	Review/transgenics	Lactose as osmotic regulator of milk volume	F8
21688337	Mouse genetics (STAT5)	LA expression under lactation-hormone control	F8
168914	Classical biochemistry	A protein / B protein nomenclature	F9
1204653	Biochemistry	B4GALT1 = A protein of lactose synthase	F9
11883930	Mechanistic review	"Molecular mechanical device"; conformation II	F9
8816760	Comparative enzymology	LA re-specifies snail β4-GalNAcT >100× toward Glc	F9
10813835	Mutagenesis/metal binding	Strong Zn^{2+} site: Glu1, Glu7, Asp11, Asp37 (bovine)	F10
9537992	Crystallography	Secondary Ca^{2+} site near functional surface	F10
7945790	Spectroscopy	Secondary Zn^{2+}/Co^{2+} binding, aggregation	F10

The evidence base is strongly weighted toward **precise, mechanistic studies** (high-resolution crystallography and classical enzyme kinetics) rather than high-throughput data, satisfying the mandate to prioritize precise over high-throughput findings. There are no contradictory findings among the reviewed literature; the enzymological, structural, evolutionary, and cell-biological lines of evidence are mutually reinforcing.

Gene/Protein Identity Verification

All verification steps are satisfied. The gene symbol **LALBA** corresponds exactly to the protein described by UniProt P00711 — bovine α-lactalbumin / Lactose synthase B protein / allergen Bos d 4. The literature uniformly concerns the correct organism (*Bos taurus*, with homologous work on human, goat, and guinea-pig α-lactalbumins used supportively and clearly labeled as such). The protein family (glycosyl hydrolase 22 / c-type lysozyme-like fold) and key domains align precisely with the published structural and evolutionary studies. There is **no ambiguity**: α-lactalbumin is a well-characterized protein, and the literature consistently describes a single coherent functional entity.

Limitations and Knowledge Gaps

1. **Bovine-specific structural data are partly inferred from homologs.** Several of the most detailed mechanistic crystal structures of the lactose synthase complex used the bovine B4GALT1 catalytic domain but human or chimeric α-lactalbumin, or were solved for human α-lactalbumin (e.g., the 1.7 Å structure, PMID: 1920433). Bovine-specific metal-binding work (PMID: 10813835) explicitly cautions that **sequence homology to human α-lactalbumin does not reliably predict subtle functional features**, so some fine details may differ between species.
2. **Precise binding-interface residues on bovine α-lactalbumin.** While the flexible C-terminal region and a functionally critical tryptophan are implicated, a complete residue-level map of the bovine α-lactalbumin / B4GALT1 interface, validated by mutagenesis in the bovine protein specifically, is not fully established in the reviewed literature.
3. **Physiological role (if any) of secondary metal sites.** The proposal that secondary Ca^{2+}/Mn^{2+} sites participate in lactose synthase complex assembly (PMID: 9537992) is a structure-based inference that has not been directly tested functionally.
4. **Quantitative in vivo flux control.** Although α-lactalbumin is clearly rate-determining for lactose synthesis, the reviewed evidence is largely qualitative/transgenic; precise quantitative control-coefficient measurements in bovine mammary tissue are not detailed here.

5. **HAMLET/BAMLET physiological relevance.** The tumoricidal activity is well documented in vitro and in models but is a non-canonical, conformation-dependent activity; its relevance (if any) to normal bovine physiology is unestablished.

Proposed Follow-up Experiments / Actions

1. **Solve or refine the fully bovine lactose synthase complex structure.** Co-crystallize *Bos taurus* B4GALT1 with *Bos taurus* α-lactalbumin (P00711), UDP-Gal, glucose, and Mn^{2+}/Ca^{2+} to map the bovine-specific interface at high resolution and resolve any species differences flagged by PMID: 10813835.
2. **Alanine/mutational scan of the bovine interface.** Systematically mutate predicted interface residues (flexible C-terminal region, the functionally critical tryptophan) in recombinant bovine α-lactalbumin and measure effects on lactose synthase k_cat/K_m to validate the specifier mechanism in the native protein.
3. **Functional test of secondary metal sites.** Mutate the proposed strong Zn^{2+}-site residues (Glu1, Glu7, Asp11, Asp37) and the secondary Ca^{2+} site, then assay complex formation and lactose synthase activity to test the structural hypothesis from PMID: 9537992.
4. **Quantitative metabolic flux analysis.** Use isotope-labeled glucose/UDP-galactose in primary bovine mammary epithelial cultures with titratable α-lactalbumin expression to quantify its control coefficient over lactose flux and milk-volume determinants.
5. **Hypoallergenic engineering.** Leverage the mapped Bos d 4 IgE epitopes (PMID: 27273451, PMID: 27603546) to design epitope-disrupted variants, testing whether specifier function can be retained while reducing allergenicity — relevant to dairy-product design.

Conclusion

Bovine α-lactalbumin (LALBA, P00711) is definitively characterized as the **non-catalytic regulatory specifier subunit of lactose synthase (EC 2.4.1.22)**. It binds β-1,4-galactosyltransferase-1, stabilizes the enzyme's closed conformation, and holds glucose in the acceptor site, thereby switching the transferase's acceptor specificity from GlcNAc to glucose so that the complex synthesizes **lactose** from UDP-galactose and glucose. This occurs in the **Golgi**

lumen of lactating mammary epithelial cells, after which the protein is **secreted into milk** as an abundant calcium-binding whey protein and the major allergen **Bos d 4**. Structurally it is a **catalytically inactive c-type lysozyme homolog** (glycosyl hydrolase family 22) and the archetypal molten-globule model protein; the lactose it enables is the **osmotic driver of milk volume**, and its hormonally restricted expression makes it the tissue-specific switch for milk-sugar production.
